

Feasibility of Mapping Engagement Ratios to Levels of Task Complexity within VR Environments

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ABSTRACT

This paper studied the feasibility of mapping an engagement ratio onto a level of task complexity, when human participants undertook interactive tasks within a virtual reality (VR) environment. Each human participant used a haptic device to push a ball-shaped object through a pipe. There were a total of three pipes, which had three different shapes corresponding to the levels of task complexity mathematically. An electroencephalogram (EEG) device recorded the brain activity of the participant while undertaking the task. The outcomes of the study confirmed the feasibility of mapping the engagement ratio with the levels of task complexity.

Keywords: VR, task complexity, EEG, engagement ratio.

Index Terms: H.1.2 [User/Machine Systems]: Human information processing; H.5.1 [Multimedia Information Systems]: Artificial, augmented, and virtual realities; H.5.2 [User Interfaces]: Haptic I/O

1 INTRODUCTION

Towards an adaptative automation capable of varying intervention degrees based on levels of task complexity [1, 2], user brain activities are measured in real time for assessing user mental states and situation awareness. The measurement needs to record the brain activities through an electroencephalogram (EEG) device as electrical signals and to process the signals for extracting certain measurement parameters – for example, an engagement ratio, ER . As shown in Eq. (1), ER is related to the power spectrum densities (PSD) of the signals within three frequency bands as [3, 4]:

$$ER = \left(\frac{\beta}{\alpha + \theta} \right), \quad (1)$$

where the parameters θ , α , and β represent the power spectrum densities within three frequency bands of 4–8 Hz, 9–12 Hz, and 13–30 Hz, respectively [3].

Using user mental states is not a new concept for adaptative and cognitive automation. However, existing reports had several drawbacks. For example, the mental state is subjectively measured as Mental Work Load to correspond several levels of task complexity [3]. The PSD-related computation of EEG signals is limited to that of the α band, causing insufficient use of the computation for different levels of task complexity [4]. A lack of adequate definitions of task complexity diminishes the usefulness of possible correlations between EEG signals and ratios of engagement and attention [5]. Moreover, most of the reports describe offline analyses of EEG signals, reducing their potential applications to an adaptative automation in real time.

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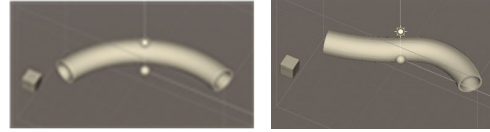


Figure 1: Two pipes with two different levels of task complexity.

Towards real-time adaptive automation, we undertook a study to correlate the engagement ratio, as indicated in Eq. (1), with the levels of task complexity. Firstly, we defined mathematically indices of task complexity. Secondly, we computed the indices to select three different levels of the complexity based on the positive increases of the indices. Lastly, we recorded EEG signals of human participants to confirm the feasibility of mapping the engagement ratio with the levels of task complexity.

2 METHODS

VR setup: Our VR environment used a Pimax 5K head mounted device (HMD) (Pimax Technology, China) to display a visual scene. The scene consisted of a pipe, which maintained its length but varied its curvature, angle of the entrance, diameter, and number of curves. Two examples of the pipe are given in Fig. 1 to represent two different levels of task complexity.

As shown in Fig. 2, each human participant wore the cap of a 20-channel Enobio EEG device (Neuroelectronics, Spain) and used his/her right hand to manipulate the stylus of a TOUCH haptic device (3D Systems, USA). To increase the comfort of the participant, two front electrodes (FP1 and FP2) of the EEG device were taken off to reduce the pressing of the HMD onto the front head just above the eyes. The other 18 electrodes were positioned according to the 10-20 system on the scalp of the participant. EEG signals of these 18 electrodes were recorded at 500Hz.

The participant used the tip of the stylus to push a ball-shaped object through the pipe – from its right end to its left counterpart. By using the tip to touch a cube near the pipe, the participant initiated his/her interactive task. A force of 0.8 N was fed back to the right hand of the participant when the ball-shaped object impacted the internal wall of the pipe. This amount of force is above the threshold (0.45 N) of human force sensing [6].

For the interactive task, the index of task complexity, TC , was defined as follows:

$$TC = L_p * \sum (Cl * (R_p - R_b) * \sin A_c), \quad (2)$$



Figure 2: The VR setup.

Table 1: Averaged measurements per level of task complexity.

Task complexity	Engagement ratio	No. of contacts	No. of completed trials	Time of a trial (s)
Baseline	0.459±0.001	-	-	-
L1	0.504±0.005	35	12	14.56
L2	0.553±0.010	15	15	12.40
L3	0.589±0.027	21	21	7.68

where L_p is the length of a pipe; Cl describes the curved length of a segment of the pipe; R_p and R_b are the radius of the segment and the ball-shaped object, respectively; and A_c gives the degree of the curved segment. TC is compounded for multiple curved segments with varied radius of the segments and ball-shaped objects, when considering the fixed length of the pipe.

For the study, we manipulated only the number of curved segments and their degrees, while unchanged the radius of the curved segments (1.0 cm), the radius of the ball-shaped object (0.5 cm), and the length of the pipe (16.0 cm). The number of the segments were from 0 to 3, and the degrees of the segments differed as -90.0° , -45.0° , 0.0° , 45.0° , 90.0° , and 180.0° . These parameters yielded a total of 10 pipes, with their indices of task complexity varying from -23.5 to 31.9 based on Eq. (2). Among these indices, six pipes with positive indices grouped into 3 pairs, which corresponded to 3 levels of task complexity at about 10.0 (L1), 23.0 (L2), and 31.0 (L3). These 3 levels were selected for testing human participants for the study.

The VR environment was developed by using Unity (2017.30f LTS) and OpenHaptics (ver 3.5), with multithreads of the visual scene, haptic interfacing, EEG recording, signal processing, and PSD computation. The programming in C# was for interaction with the visual scene, in C++ for thread handling, and in Python for customizing the processing and PSD computation of EEG signals. The VR environment ran on a graphic computer with an Intel Xeon E5400 CPU, 64 GB RAM, an Nvidia Quadro RTX5000 GPU, and the Windows 10 operating system.

Participants: Ten healthy volunteers (3 females and 7 males, aged at 26.7 ± 1.7 years) were recruited to participate in the study. The number of the participants was larger than a necessary minimum according to Lehr's formula [7]. Being naïve to the purpose of the study, each participant successfully passed a pre-test to confirm his/her right-handedness, normal color vision with stereo acuity of at least $40''$ of arc, and no physical impairment of moving his/her right hand. The study had an ethics clearance from our institute. All participants gave their informed consent for their participation in the study.

Procedure: Taking a seat comfortably, each participant used his/her right hand to hold the stylus of the haptic device while wearing the HMD and EEG cap. The participant underwent a baseline, a learning, and a testing session. With the protocol of eye-closed-then-eye-open and fixing attention to a cross centred on the HMD, the baseline session lasted 3.0 minutes to record the participant's rest state of EEG signals. Then, the participant underwent the learning session for 5.0 minutes to ensure that he/she could consistently carried out the interactive task with less than 40 contacts between the object and the internal wall of the pipe. Finally, the participant undertook the testing session that had 30 trials, 10 trials per level of task complexity. The order of the 3 levels was always from L1, L2, to L3 for the participant. The session lasted 30 ~ 45 minutes per participant.

For each participant, data recording was EEG raw signals and the behavioural metrics including: the number of contacts between the object and the pipe's internal wall, the number of trials per level completed within 3.0 minutes, and the completion time of a trial.

A Python algorithm was developed to process the stream of EEG signals acquired from the EEG device in real time. The process performed filtering to remove AC line artifacts at 60Hz and computed finite impulse response (FIR, 1 ~ 30hz) to eliminate the effect of muscular impulses. Then, the computation of PSD used Welch periodogram over a window of 2.0 seconds. The filtering and PSD computation were carried out over all channels except FP1 and FP2; and averaged over all trials that the participant completed. The time of both the filtering and computation was about 0.57 seconds, considering as real time.

Each of the participants spent about 2.00 hours for the study, including the pre-test, three sessions (the baseline, the learning, and the testing), and breaks.

3 RESULTS AND DISCUSSION

Averagely, each participant undertook about 72 trials. Table 1 gives average measurements per level of task complexity over the participants. As expected, the engagement ratio (ER) increased to correspond to the increased level of task complexity. However, this increase was not uniform across all levels. From L1 to L2, the increase was more evident than that from L2 to L3. Moreover, L3 had a larger variance than the other two levels. Because ER of L2 was not normally distributed among the participants, a one-way ANOVA analysis of ER was carried out among the baseline, L1, and L3. The analysis indicated a significant difference of ER [$F(2, 27) = 86.31$; $p < 0.001$]. Post-hoc Bonferroni tests confirmed that the difference was significant ($p < 0.001$) for each pair among the baseline, L1, and L3.

Complementing to ER , two distinct groups of the participants exhibited different behavioral metrics. One group contacted the internal wall of the pipe more frequently but performed the task faster than another group. Interestingly, the number of contacts decreased when the level of task complexity increased, as shown in Table 1. This might result from the fixed order of the levels in testing, enabling the participants to get used with the task over time.

4 CONCLUSION

This study indicates that it is feasible to differentiate specific levels of task complexity based on ER for an interactive task. Future work will examine behavioural metrics against levels of task complexity.

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